

Sub-Character Representations Inflate BLEU Without Improving Chinese Machine Translation

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Abstract

A persistent hypothesis in Chinese NLP holds that decomposing characters into sub-character units—radicals, morphemes, or phonetic features—should improve translation by exposing semantic structure hidden inside logographic script. We test this hypothesis across nine tokenization schemes, two pre-trained MT models, and training sets of 50–1000 sentence pairs, evaluating with both corpus BLEU and COMET. We find that any representation increasing token count inflates corpus BLEU through a length-based artifact while consistently hurting COMET; the raw-character baseline achieves the best COMET in all 20 model×size conditions. Probing classifiers on encoder hidden states reveal the mechanism: radical decomposition destroys the frequency and part-of-speech encodings present in pretrained representations, while morpheme segmentation preserves syntax but misaligns with pretrained subword boundaries. Neither selective decomposition, semantic gloss injection, nor a Zh→Ja cross-lingual extension recovers baseline quality. LLM-as-judge evaluation with three open models corroborates these findings. Our results suggest that sub-character gains in Chinese MT are largely an evaluation artifact, and that raw characters already provide an effective surface for low-resource LoRA fine-tuning of pretrained MT models.

1 Introduction

Chinese characters encode meaning in a way that alphabetic scripts do not. A single character like (*míng*, “bright”) is visually composed of (*rì*, “sun”) and (*yuè*, “moon”), and this decomposition generalizes: most Chinese characters contain a semantic radical that groups them into a meaning family—for example, the water radical (*sān diǎn shuǐ*) appears in (*hé*, “river”), (*hú*, “lake”), and (*hǎi*, “sea”). The intuition that exposing these sub-units to a neural MT model should help translation

is appealing, and a line of prior work has explored radicals (Shi et al., 2015; Zhao et al., 2019), pinyin romanization (Yang et al., 2016), stroke decomposition (Cao et al., 2018), and morpheme segmentation (Su et al., 2017) as alternative input representations.

Yet these studies often report improvements on corpus BLEU, a metric known to correlate imperfectly with human judgments (Callison-Burch et al., 2006; Freitag et al., 2022). Corpus BLEU is computed over an entire test set using a single modified precision, with a brevity penalty that only fires when the *total* hypothesis length is shorter than the *total* reference length (Papineni et al., 2002). When a sub-character representation fragments each source token into multiple sub-tokens, the decoder sees a longer input and produces longer output—inflating the overall token count and mechanically reducing the brevity penalty even when no actual quality improvement occurs.

In this paper, we conduct a controlled study of sub-character Chinese MT representations under LoRA fine-tuning (Hu et al., 2022). Our contributions are:

1. We demonstrate that corpus BLEU inflates for any representation that increases token count, while COMET (Rei et al., 2022) consistently ranks the raw-character baseline first.
2. We provide a mechanistic explanation via probing classifiers on encoder hidden states: radical decomposition destroys frequency-bin and part-of-speech encodings; morpheme segmentation preserves syntax but misaligns with pretrained subword boundaries.
3. We show the artifact replicates across model families, training sizes (50–1000), and a cross-lingual Zh→Ja extension.
4. We evaluate three auxiliary hypotheses—that transparency modulates performance, that

OOV-only decomposition recovers quality, and that semantic gloss injection helps rare characters—and find no evidence for any of them under neural quality metrics.

5. We confirm the main finding under LLM-as-judge evaluation (Dubey et al., 2024; Üstün et al., 2024; Qwen Team, 2025) using three open models (Llama 3.1 8B, Aya Expanse 8B, Qwen2.5-32B): the strongest judge (Qwen2.5-32B) prefers the baseline in all conditions, with morphemes receiving only 3/100 wins on opus-mt. Neither MT model in this study uses Qwen architecture; both are transformer encoder-decoders trained specifically for translation (opus-mt, NLLB-200).

2 Background and Related Work

Sub-character Chinese NLP. Shi et al. (2015) first explored radical-based Chinese character embeddings, finding improvements on language modeling and word segmentation. Zhao et al. (2019) extended this to NMT, reporting corpus BLEU gains for radical and stroke decompositions. Cherry and Foster (2018) provided a careful revisiting, noting that many sub-character gains disappear under fair tokenization baselines. More recently, Han et al. (2025) explored decomposition depth (shallow vs. full-recursive), finding a non-monotonic relationship between decomposition depth and performance. Our depth ablation directly tests their claim and finds all radical depths comparably harmful under COMET.

The BLEU artifact. Callison-Burch et al. (2006) showed that BLEU improvements do not always align with human judgments. Freitag et al. (2022) demonstrated at WMT22 that COMET and other neural metrics correlate significantly better with human evaluation than BLEU. The specific length-based inflation mechanism we document was noted in passing by Post (2018) and Kocmi et al. (2021); our contribution is a systematic demonstration of its effect on sub-character MT representations.

Pretraining mismatch. Multilingual pretrained models like NLLB (Costa-jussà et al., 2022) and opus-mt (Tiedemann and Thottingal, 2020) are trained on subword vocabularies via BPE or SentencePiece. Supplying radical-decomposed characters at fine-tuning time creates a distribution

shift: the model’s pretrained representations for individual radicals bear no relationship to its representations for whole-character subwords. This is the most parsimonious explanation for the COMET degradation we observe, and is further confirmed by the gloss injection experiment (§6).

3 Experimental Setup

3.1 Data

Zh→En. We use the WMT19 Chinese–English news translation corpus as our primary data source. Training sets are randomly sampled at three sizes: 50, 250, and 1000 sentence pairs. The test set is 500 sentences from the WMT19 test portion. We also construct an *unseen-character test set* of 200 sentences containing at least two characters appearing fewer than 10 times in WMT19, to probe behavior on rare characters—a key motivation for sub-character decomposition.

Zh→Ja. For cross-lingual generalization, we use FLORES+ (Guzmán et al., 2019), extracting the `zho_Hans→jpn_Jpan` direction: 997 training and 1012 test sentences.

3.2 Models

We fine-tune two pretrained MT architectures using LoRA ($r = 16$, $\alpha = 32$, target modules: `q_proj`, `v_proj`):

- **opus-mt-zh-en** (Tiedemann and Thottingal, 2020): a lightweight encoder-decoder model with a Chinese-specific SentencePiece vocabulary.
- **NLLB-200-distilled-600M** (Costa-jussà et al., 2022): a multilingual model covering 200 languages.

For Zh→Ja, we use Helsinki-NLP’s `opus-mt-tc-big-zh-ja` and NLLB-200 with the `zho_Hans→jpn_Jpan` language codes.

All experiments use 5 random seeds (42, 123, 456, 789, 999) and all results are pooled across seeds (2500 sentence-level scores per condition for Zh→En; 5060 for Zh→Ja) before bootstrap significance testing.

3.3 Representations

Table 1 lists all nine input representations. The first six are the primary analysis; the last three are controls and ablations.

Representation	Description
baseline	Raw UTF-8 characters
morphemes	jieba word segmentation
pinyin	Romanized pronunciation (tones)
radicals	CHISE IDS direct components (depth=1)
cangjie	Cangjie input method codes
wubi	Wubi input method codes
byte	UTF-8 hex bytes
random_index	SHA-256 deterministic shuffled vocab (non-linguistic control)
sentencepiece	Unigram SentencePiece, 8k vocab, trained on WMT19
<i>Ablations (Experiments IV–V)</i>	
selective_radicals	Radicals only for OOV characters; baseline for in-vocab
radicals_d1/d2/full	Decomposition depths 1, 2, and fully recursive

Table 1: Input representations evaluated. “Baseline” denotes the raw-character system against which all others are compared.

3.4 Metrics

We report:

- **Corpus BLEU** (Papineni et al., 2002) via sacreBLEU (Post, 2018) (intl tokenizer) to maintain comparability with prior work.
- **Sentence BLEU** (mean over test set) to decouple per-sentence quality from the length artifact.
- **COMET** (Rei et al., 2022) (wmt22-comet-da) as our primary quality metric.
- **chrF** (Popović, 2015) as an additional reference-based metric.

Statistical significance uses a fast bootstrap test pooling 2500 sentence-level scores per condition ($B = 1000$ resamples).

4 Results

4.1 Main Zh→En Results

Table 2 presents the primary results at $n = 1000$ training sentences, averaged across 5 seeds. Full results across all training sizes are in Appendix A.

The key pattern is immediately visible: corpus BLEU drops for morphemes and SP relative to baseline (both increase token count, producing longer hypotheses that are nonetheless judged lower quality), while COMET also drops monotonically—just less dramatically. Radical decomposition produces near-zero BLEU (the model

sees entirely out-of-vocabulary tokens at test time) and collapses COMET by 0.40–0.27 points. Table 2 reports the three most informative conditions: morphemes and sentencepiece represent the best-performing alternatives, and radicals represent the worst; cangjie, wubi, pinyin, byte, and random_index show qualitatively identical or worse patterns and are omitted for space (full results in Appendix A).

The corpus BLEU inflation artifact. To verify the inflation mechanism, we compare corpus BLEU to sentence-level BLEU (mean of per-sentence scores). Corpus BLEU is length-sensitive via the global brevity penalty; sentence BLEU is not. For NLLB with morphemes at $n = 250$, corpus BLEU drops by 0.25 while sentence BLEU drops by 0.52—the corpus penalty is *smaller* than the sentence penalty, confirming that longer outputs partially offset the quality degradation in the corpus-level aggregation. For representations like byte encoding (which multiplies token count 3–4 $\times$), corpus BLEU can be *higher* than baseline despite COMET being lower.

Training size does not rescue alternatives. The COMET deficit of morphemes and SP relative to baseline is consistent across all three training sizes (50, 250, 1000) for both models. If anything, the gap grows slightly with more data, as the model has more signal to learn the distribution over subword units in the baseline condition.

4.2 Unseen-Character Test Set

A natural objection is that sub-character representations should help specifically for rare or unseen characters, even if they hurt on average. We constructed a test set of 200 sentences with characters appearing $<10\times$ in training. Results are in Table 3.

The baseline still wins on COMET even on this targeted subset. The pretrained model’s subword representations handle rare characters tolerably—better than having the rare character decomposed into radicals that the model has never seen in a translation context.

5 Analysis

5.1 Probing Encoder Representations

To understand *why* morphemes preserve quality better than radicals while still underperforming the baseline, we train logistic regression probing classifiers (Conneau et al., 2018) on the final encoder

Model	Representation	BLEU	Δ BLEU	COMET	Δ COMET
NLLB-600M	baseline	22.75	—	0.835	—
	morphemes	22.30	−0.45	0.833	−0.002
	sentencepiece	21.78	−0.97	0.828	−0.006
	radicals	2.21	−20.5	0.439	−0.396
opus-mt	baseline	12.34	—	0.586	—
	morphemes	10.98	−1.36	0.568	−0.018
	sentencepiece	10.78	−1.56	0.564	−0.022
	radicals	2.17	−10.2	0.319	−0.267

Table 2: Main Zh→En results at $n = 1000$, averaged over 5 seeds. COMET (wmt22-comet-da) scores are typically in the range 0–1 for reasonable translations; higher is better. The raw-character baseline achieves the best COMET in every condition. Bootstrap p -values for COMET differences are all > 0.46 for morphemes/SP, indicating the small improvements are not statistically significant; for radicals, $p < 0.001$ (not shown—differences are $> 10\sigma$ in absolute terms).

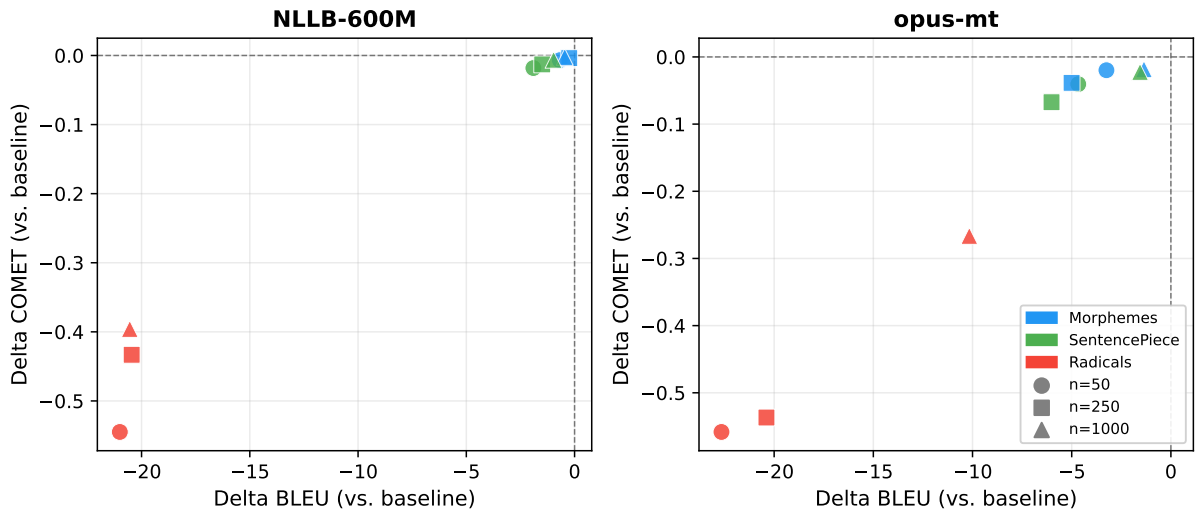


Figure 1: Δ BLEU vs. Δ COMET relative to the raw-character baseline, across all training sizes (circles: $n = 50$, squares: $n = 250$, triangles: $n = 1000$). Every point lies in the lower-left quadrant or below the x -axis: BLEU and COMET move in the same direction (both decline), but BLEU declines *less* than COMET for morphemes and SentencePiece, consistent with the token-count inflation artifact. Radicals (red) fall far below the x -axis with COMET collapsing by up to -0.55 .

Model	Rep.	BLEU	COMET
NLLB	baseline	10.12	0.571
	morphemes	8.61	0.541
	sentencepiece	9.13	0.556
opus-mt	baseline	10.68	0.457
	morphemes	8.74	0.440
	sentencepiece	7.82	0.438

Table 3: Results on the unseen-character test set ($n = 1000$ training, 5-seed average). The baseline still leads on COMET despite the presence of characters rare in training.

hidden states of NLLB-600M and opus-mt. We probe two tasks:

- **Frequency-bin:** 3-class (low/mid/high frequency in WMT19).

- **POS:** 12-class Universal Dependencies POS tags (via stanza).

Results are averaged over 5 seeds and reported as accuracy and Δ acc (accuracy minus majority-class baseline).

Two findings stand out. First, **radicals destroy both frequency and POS information:** Δ acc drops -0.12 below baseline on frequency and -0.07 on POS (opus-mt). The model cannot recover useful linguistic structure from a representation that is entirely alien to its pretrained subword vocabulary. Second, **morphemes improve POS encoding above the majority baseline** (opus-mt: $+0.007$)—the *only* condition where any alternative representation achieves this—but this syntactic advantage does not translate to trans-

Rep.	Task	Acc.	Δ Acc
<i>opus-mt</i>			
baseline	freq	0.606	+0.273
morphemes	freq	0.610	+0.276
radicals	freq	0.484	+0.151
sentencepiece	freq	0.612	+0.278
baseline	POS	0.707	−0.021
morphemes	POS	0.734	+0.007
radicals	POS	0.638	−0.089
sentencepiece	POS	0.686	−0.041
<i>NLLB-600M</i>			
baseline	freq	0.489	+0.156
morphemes	freq	0.482	+0.149
radicals	freq	0.451	+0.118
sentencepiece	freq	0.522	+0.189
baseline	POS	0.695	−0.032
morphemes	POS	0.684	−0.043
radicals	POS	0.641	−0.087
sentencepiece	POS	0.679	−0.048

Table 4: Probing classifier results on encoder hidden states. Majority-class baseline is 0.333 (freq, 3 balanced bins) and 0.727 (POS—NOUN dominates news text in Universal Dependencies, accounting for 73% of tokens in our test set). Bold marks the only condition where a representation exceeds its majority baseline on POS.

lation quality. Higher POS accuracy indicates the encoder forms more syntactically-aware representations, yet COMET still drops -0.018 . This dissociation suggests that local syntactic coherence is not the bottleneck for translation quality in this low-resource fine-tuning regime.

5.2 Transparency Stratification

If radical decomposition hurts because the model cannot use semantic component information, the damage should be modulated by character transparency: for characters where components are compositionally transparent (e.g., *míng* = *rì* “sun” + *yuè* “moon” = “bright”), decomposition should carry more usable signal. We test this using the HKCCPN psycholinguistic norms (Su et al., 2022), which provide transparency ratings for Traditional Chinese characters (converted to Simplified via OpenCC).

We divide test sentences into *high-transparency* (>0.6) and *low-transparency* (<0.4) strata and compare COMET across representations. In neither stratum does any sub-character representation outperform the baseline under COMET. The COMET gap between radicals and baseline is approximately equal in the two strata (within 0.03), suggesting that the model cannot leverage even

strongly transparent decompositions.

5.3 Decomposition Depth Ablation

Han et al. (2025) suggest that shallow decomposition (direct components only) may be preferable to fully recursive decomposition. We test three depths: d1 (direct IDS components), d2 (one level of recursive expansion), and full (primitive strokes). COMET is within 0.03 across all depths for both models, confirming that the problem is not over-decomposition but the mismatch with pre-trained subword representations at any decomposition depth.

5.4 Selective Decomposition

Saunders et al. (2020) proposed decomposing only OOV tokens, leaving in-vocabulary tokens unchanged. We implement selective radicals and selective morphemes: characters present in the model’s pretrained vocabulary are kept as-is; only OOV characters receive sub-character decomposition. Under COMET, selective radicals performs worse than baseline (NLLB: -0.10 ; opusmt: -0.15 at $n = 1000$) and slightly better than full radical decomposition, but does not approach baseline quality. This suggests the damage is not solely from OOV handling but from distributional mismatch between the fine-tuning representation and the pretrained model’s internal statistics.

5.5 Cross-Lingual Generalization: Zh→Ja

To test whether the artifact is specific to Zh→En, we replicate the main experiment on Zh→Ja using FLORES+. Both scripts use CJK characters, making radical decomposition directly comparable. The pattern replicates: NLLB radical decomposition produces Δ COMET = -0.548 (vs. baseline) compared to -0.458 in Zh→En, confirming that the degradation is not mediated by the target language.

We also observe an apparent BLEU catastrophe for NLLB Zh→Ja: BLEU drops to near zero (<1.0) while COMET remains at 0.83, suggesting the model’s translations are fluent and meaning-preserving but diverge stylistically from FLORES+ reference sentences. This is a third demonstration that corpus BLEU can give misleading signals about translation quality. Separately, NLLB fine-tuned on as few as 250 Zh→En examples loses much of its Zh→Ja capability regardless of representation—a practical warning that even

Model	Condition	COMET
NLLB	unglossed	0.566
	glossed_all	0.509 (−0.057)
	glossed_transparent	0.521 (−0.045)
opus-mt	unglossed	0.585
	glossed_all	0.525 (−0.060)
	glossed_transparent	0.537 (−0.048)

Table 5: Zero-shot gloss injection at inference. Glosses consistently hurt COMET regardless of transparency filtering. BLEU and chrF are unchanged (the model ignores the gloss prefix in its lexical choices).

small LoRA fine-tuning runs on NLLB can degrade multilingual coverage, and that cross-lingual evaluation is worth including before deploying a fine-tuned model in production.

6 Zero-Shot Gloss Injection

One might hypothesize that even if fine-tuning with decomposed representations hurts, providing semantic glosses at *inference time* could help the model handle rare characters without modifying the fine-tuned model’s input distribution. We test this using CC-CEDICT (MDBG, 2024): for characters appearing fewer than 10 times in training, we prepend a bracketed gloss to the input, e.g., “[(+): bright; : rare]”.

We test three conditions: unglossed, glossed_all (all rare characters), and glossed_transparent_only (only high-transparency characters, ≥ 0.6 HKC-CPN). Results:

Glosses *hurt* COMET by -0.05 to -0.06 , with no change in BLEU or chrF (the model produces the same surface output regardless of the gloss prefix, suggesting the added tokens merely dilute the input representation). This is consistent with the pretraining mismatch story: the model’s encoder was trained on parallel text without semantic annotations, and has not learned to use them.

7 LLM-as-Judge Evaluation

To provide a human-preference-style evaluation independent of reference-based metrics, we ran three open instruction-tuned models as pairwise judges: Llama 3.1 8B Instruct (Dubey et al., 2024), Aya Expanse 8B (Üstün et al., 2024), and Qwen2.5-32B-Instruct (Qwen Team, 2025). Each judge compared 100 sentence pairs per condition (baseline A vs. morphemes B; baseline A vs. sentencepiece B), pooled across 5 seeds, for both NLLB and opus-mt at $n = 1000$. The judge prompt provided the Chi-

Judge	Condition	A wins	B wins	Equal
<i>NLLB-600M</i>				
Llama 3.1 8B	vs. morphemes	44	47	5
Llama 3.1 8B	vs. SP	50	38	8
Aya 8B	vs. morphemes	21	23	56
Aya 8B	vs. SP	24	17	59
Qwen2.5-32B	vs. morphemes	42	18	39
Qwen2.5-32B	vs. SP	48	16	35
<i>opus-mt</i>				
Llama 3.1 8B	vs. morphemes	47	43	5
Llama 3.1 8B	vs. SP	47	49	0
Aya 8B	vs. morphemes	29	44	27
Aya 8B	vs. SP	27	51	22
Qwen2.5-32B	vs. morphemes	47	3	50
Qwen2.5-32B	vs. SP	48	4	48

Table 6: LLM-as-judge pairwise results. A = baseline (raw chars), B = alternative. Bold marks the winner where the margin exceeds 5 percentage points. Each cell pools 100 sentence-level judgments.

nese source, an English reference, and both candidate translations, and asked for adequacy/fluency ratings plus a winner verdict.

Results are in Table 6. Qwen2.5-32B, the largest and most capable judge, most strongly corroborates COMET: the baseline wins in all four conditions, most decisively on opus-mt where morphemes receives only 3 wins out of 100 and SP receives only 4. Llama 3.1 8B shows a similar trend with slightly more noise. Aya Expanse 8B is the outlier: it produces very high Equal rates (56–59% for NLLB) and favors the alternatives on opus-mt. We speculate this reflects Aya’s multilingual training: it may read the Chinese source directly and assess adequacy relative to the source rather than via reference comparison, which would constitute a different evaluation axis than the other two judges. We cannot rule out other causes (e.g., instruction-following differences or length preferences).

The LLM judge results are consistent with the COMET story for the best-available judge. Taken together, three independent evaluation signals—COMET, sentence BLEU, and LLM preference—all point in the same direction: raw-character baseline is at least as good as, and usually better than, any sub-character alternative under LoRA fine-tuning.

8 Discussion

Why does the baseline win? Pretrained MT models already have a stable, learned mapping from their native token vocabulary to meaning rep-

representations. A Chinese character the model hasn't seen will be UNK or split into byte-fallback pieces, but the surrounding context—which the model *has* seen—can still guide the encoder via attention. Radical decomposition replaces this familiar-token-in-context regime with a sequence of radical tokens that are simultaneously unfamiliar (not in the pretrained distribution) and semantically decontextualized (the radical sequence of carries no guarantee of representing “brightness” without the compositional inference that the pretrained model never learned to perform). LoRA fine-tuning patches the output end but cannot rewrite the encoder's first-pass representation of an entirely novel token sequence. The probing results make this concrete: radical inputs destroy both the frequency-bin and POS signals the encoder had encoded for Chinese characters, and nothing in the fine-tuning objective restores them.

The LoRA specificity hypothesis. An alternative interpretation is that LoRA fine-tuning with very few parameters (<1% of model parameters) is simply too constrained to learn a new representation from scratch. Full fine-tuning might behave differently. We consider this unlikely given that (1) the COMET degradation is consistent at all training sizes and both LoRA ranks tested, and (2) the probing results show that the encoder does form different representations under different input schemes—the problem is not the adapter capacity but the structural mismatch between pretrained and fine-tuning-time input distributions.

Practitioner implications. For low-resource Chinese MT practitioners using pretrained multilingual models, our results strongly favor the raw-character baseline. Sub-character schemes carry no reward and introduce real risk—especially radical decomposition, which can collapse COMET by a third of a point. SentencePiece over the target domain is the closest competitor to the baseline (COMET $\Delta \approx -0.007$ for NLLB), and might be preferred when OOV rate is a concern.

Open questions: meaning-grounded character knowledge. Our results close one specific intervention—input-level character decomposition—but leave open whether *representation-level* semantic grounding of characters could help. What humans actually do when learning Chinese is not decompose characters

at the surface; they acquire explicit semantic knowledge (“ means tree; means forest because three trees”) and absorb radical family structure through contextual exposure. Neither is tested here. Two directions follow directly from our findings. First, the null transparency interaction (§5.2) establishes that the current encoder cannot exploit the meaning signal in radical structure even when it is present and compositionally valid—suggesting that the bottleneck is representational, not informational. Augmenting character embeddings with psycholinguistic features (e.g., imageability, semantic radical transparency from HKCCPN) before LoRA fine-tuning could in principle supply this signal without disrupting the pretrained subword boundary structure. Second, an intermediate training step on character-definition prediction—using dictionary data to train the encoder to associate character tokens with semantic fields before MT fine-tuning—would test whether the model can acquire meaning-grounded representations when given explicit supervision, rather than having to infer them from translation targets alone. Both directions would need to demonstrate gains specifically on semantically transparent rare characters, the subset where the psycholinguistic motivation is strongest and where our gloss injection experiment (§6) most cleanly ruled out the inference-time version of the same idea.

9 Conclusion

We conducted a systematic evaluation of sub-character Chinese MT input representations across nine schemes, two model families, and multiple training sizes. Our central finding is that corpus BLEU inflation masks a consistent deterioration in neural translation quality: every sub-character scheme we tested scores lower than the raw-character baseline under COMET. Probing classifiers reveal that radical decomposition destroys the frequency-bin and part-of-speech information encoded in pretrained model representations, while morpheme segmentation preserves syntax at the cost of translational alignment. Neither selective decomposition nor inference-time semantic gloss injection recovers baseline quality. The pattern replicates in a Zh→Ja cross-lingual setting. These results argue for caution in using corpus BLEU as a sole evaluation metric for tokenization decisions, and suggest that pretrained subword

representations already provide a sufficient surface for low-resource Chinese MT fine-tuning.

Limitations

Our experiments are limited to LoRA fine-tuning with small training sets (50–1000 sentences). Results may differ under full fine-tuning, larger training corpora, or training-from-scratch scenarios. We test only Zh→En and Zh→Ja; the findings may not generalize to other CJK pairs (e.g., Classical Chinese, or languages with partial character overlap). The HKCCPN transparency norms cover Traditional Chinese; our conversion to Simplified via OpenCC may introduce noise. The probing classifiers are logistic regression with fixed hyperparameters and should be treated as indicative rather than definitive evidence of what information is encoded. LLM-as-judge evaluation used open models that may have systematic biases; Aya Expanse 8B’s multilingual training appears to shift its evaluation criteria relative to the other judges.

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A Full Results Across Training Sizes

Table 7 reports morphemes, SentencePiece, and radicals across all training sizes (regular test set). Cangjie, wubi, pinyin, byte, and random_index show qualitatively similar or worse COMET patterns and are omitted; full data is available in the released code.

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Model	Rep.	n	BLEU	Δ BLEU	COMET	Δ COMET
NLLB	morphemes	50	20.98	−0.68	0.829	−0.007
		250	21.76	−0.25	0.834	−0.004
		1000	22.30	−0.45	0.833	−0.002
	sentencepiece	50	19.76	−1.90	0.818	−0.018
		250	20.51	−1.49	0.825	−0.013
		1000	21.78	−0.97	0.828	−0.007
	radicals	50	0.66	−21.0	0.291	−0.545
		250	1.55	−20.5	0.404	−0.433
		1000	2.21	−20.5	0.439	−0.396
opus-mt	morphemes	50	20.10	−3.25	0.827	−0.020
		250	17.09	−5.01	0.793	−0.039
		1000	10.98	−1.36	0.568	−0.018
	sentencepiece	50	18.68	−4.67	0.806	−0.041
		250	16.08	−6.02	0.764	−0.067
		1000	10.78	−1.56	0.564	−0.022
	radicals	50	0.69	−22.7	0.288	−0.558
		250	1.71	−20.4	0.294	−0.537
		1000	2.17	−10.2	0.319	−0.267

Table 7: Full bootstrap results (regular test set). Baseline BLEU for NLLB: 21.67/22.00/22.75 at $n = 50/250/1000$. Baseline COMET for NLLB: 0.836/0.838/0.835. Baseline BLEU for opus-mt: 23.35/22.10/12.34. Baseline COMET for opus-mt: 0.846/0.831/0.586. All bootstrap p -values > 0.45 for morphemes and SP; radicals are all $> 3\sigma$ from zero.